

Respiratory

“There’s so much pollution in the air now that if it weren’t for our lungs, there’d be no place to put it all.”
—Robert Orben

“Freedom is the oxygen of the soul.”
—Moshe Dayan

“Whenever I feel blue, I start breathing again.”
—L. Frank Baum

“Life is not the amount of breaths you take; it’s the moments that take your breath away.”
—Will Smith, *Hitch*

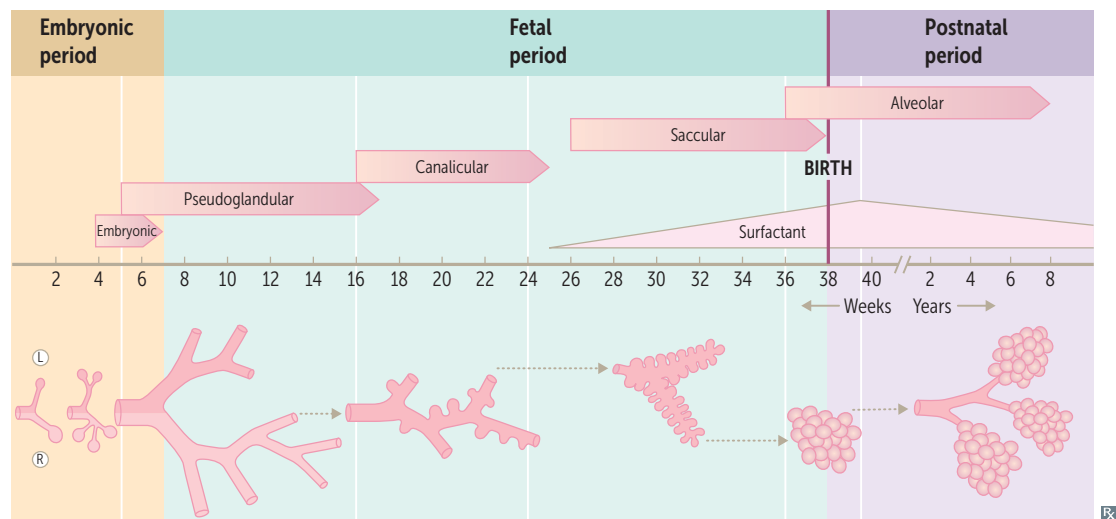
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► RESPIRATORY—EMBRYOLOGY

Lung development

Occurs in five stages. Initial development includes development of lung bud from distal end of respiratory diverticulum during week 4.

STAGE	IMPORTANT TERMS	NOTES
Embryonic (weeks 4–7)	Lung bud → trachea → bronchial buds → mainstem bronchi → secondary (lobar) bronchi → tertiary (segmental) bronchi.	Errors at this stage can lead to tracheoesophageal fistula.
Pseudoglandular (weeks 5–17)	Endodermal tubules → terminal bronchioles. Surrounded by modest capillary network.	Respiration impossible, incompatible with life.
Canalicular (weeks 16–25)	Terminal bronchioles → respiratory bronchioles → alveolar ducts. Surrounded by prominent capillary network.	Airways increase in diameter. Respiration capable at 25 weeks.
Saccular (week 26–birth)	Alveolar ducts → terminal sacs. Terminal sacs separated by 1° septae. Pneumocytes develop.	
Alveolar (week 36–8 years)	Terminal sacs → adult alveoli (due to 2° septation). In utero, “breathing” occurs via aspiration and expulsion of amniotic fluid → ↑ vascular resistance through gestation. At birth, fluid gets replaced with air → ↓ in pulmonary vascular resistance.	At birth: 20–70 million alveoli. By 8 years: 300–400 million alveoli.

**Congenital lung malformations**

Pulmonary hypoplasia	Poorly developed bronchial tree with abnormal histology. Associated with congenital diaphragmatic hernia (usually left-sided), bilateral renal agenesis (Potter sequence).
Bronchogenic cysts	Caused by abnormal budding of the foregut and dilation of terminal or large bronchi. Discrete, round, sharply defined, fluid-filled densities on CXR (air-filled if infected). Generally asymptomatic but can drain poorly, causing airway compression and/or recurrent respiratory infections.

Club cells

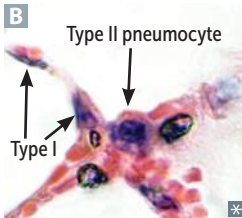
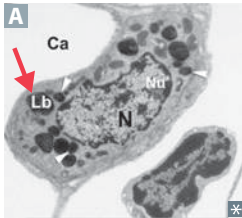
Nonciliated; low-columnar/cuboidal with secretory granules. Located in small airways. Secrete component of surfactant; degrade toxins; act as reserve cells.

Alveolar cell types**Type I pneumocytes**

97% of alveolar surfaces. Line the alveoli.
Squamous; thin for optimal gas diffusion.

Type II pneumocytes

Secrete surfactant from lamellar bodies (arrow in **A**) → ↓ alveolar surface tension, prevents alveolar collapse, ↓ lung recoil, and ↑ compliance. Cuboidal and clustered **B**. Also serve as precursors to type I cells and other type II cells. Proliferate during lung damage.

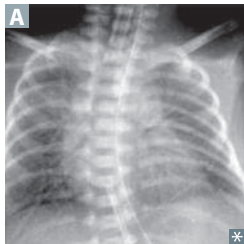
**Alveolar macrophages**

Phagocytose foreign materials, release cytokines and alveolar proteases.

$$\text{Collapsing pressure (P)} = \frac{2 (\text{surface tension})}{\text{radius}}$$

Alveoli have ↑ tendency to collapse on expiration as radius ↓ (law of Laplace).

Pulmonary surfactant is a complex mix of lecithins, the most important of which is dipalmitoylphosphatidylcholine (DPPC). Surfactant synthesis begins around week 26 of gestation, but mature levels are not achieved until around week 35.

Neonatal respiratory distress syndrome

Surfactant deficiency → ↑ surface tension → alveolar collapse (“ground-glass” appearance of lung fields) **A**.

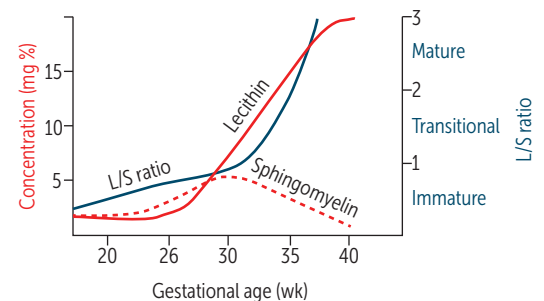
Risk factors: prematurity, maternal diabetes (due to ↑ fetal insulin), C-section delivery (↓ release of fetal glucocorticoids; less stressful than vaginal delivery).

Complications: PDA, necrotizing enterocolitis.

Treatment: maternal steroids before birth; exogenous surfactant for infant.

Therapeutic supplemental O₂ can result in **R**etinopathy of prematurity, **I**ntraventricular hemorrhage, **B**ronchopulmonary dysplasia (**RIB**).

Screening tests for fetal lung maturity: lecithin-sphingomyelin (L/S) ratio in amniotic fluid (≥ 2 is healthy; < 1.5 predictive of NRDS), foam stability index test, surfactant-albumin ratio. Persistently low O₂ tension → risk of PDA.



► RESPIRATORY—ANATOMY

Respiratory tree**Conducting zone**

Large airways consist of nose, pharynx, larynx, trachea, and bronchi. Small airways consist of bronchioles that further divide into terminal bronchioles (large numbers in parallel → least airway resistance).

Warms, humidifies, and filters air but does not participate in gas exchange → “anatomic dead space.”

Cartilage and goblet cells extend to end of bronchi.

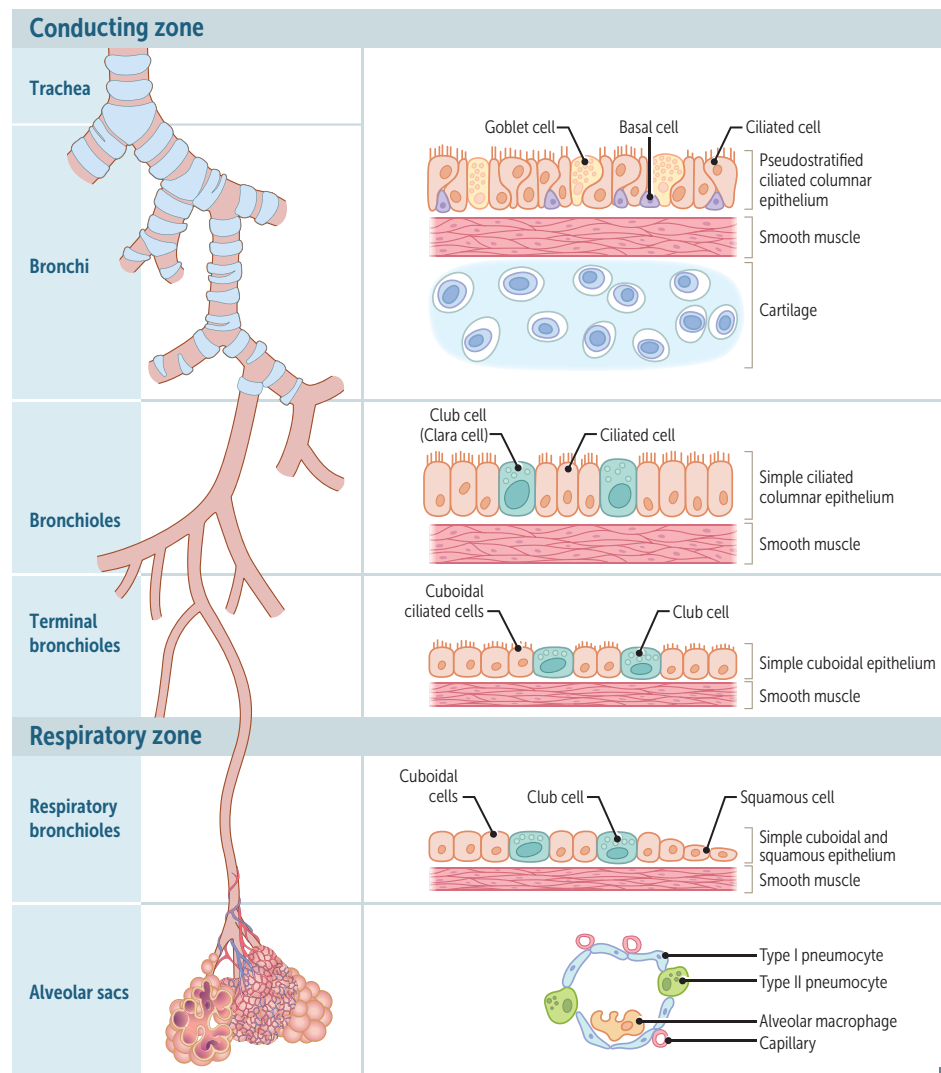
Pseudostratified ciliated columnar cells primarily make up epithelium of bronchus and extend to beginning of terminal bronchioles, then transition to cuboidal cells. Clear mucus and debris from lungs (mucociliary escalator).

Airway smooth muscle cells extend to end of terminal bronchioles (sparse beyond this point).

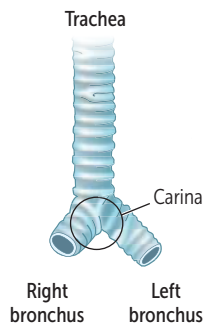
Respiratory zone

Lung parenchyma; consists of respiratory bronchioles, alveolar ducts, and alveoli. Participates in gas exchange.

Mostly cuboidal cells in respiratory bronchioles, then simple squamous cells up to alveoli. Cilia terminate in respiratory bronchioles. Alveolar macrophages clear debris and participate in immune response.



Lung relations

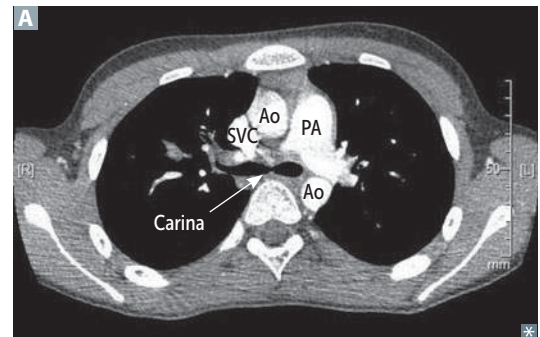
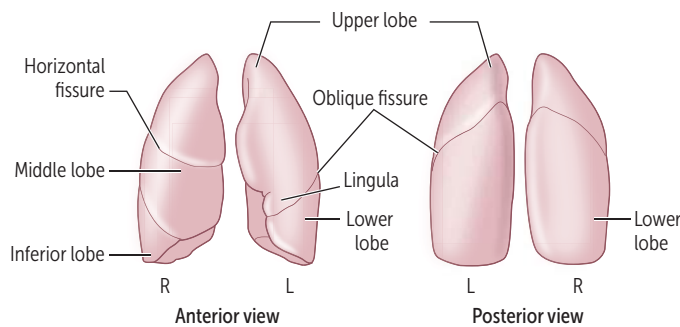


Right lung has 3 lobes; **L**eft has **L**ess **L**obes (2) and **L**ingula (homolog of right middle lobe). Instead of a middle lobe, left lung has a space occupied by the heart.

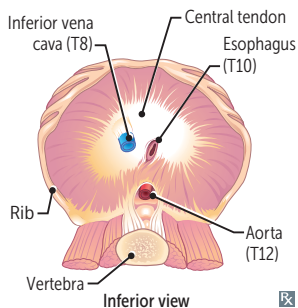
Relation of the pulmonary artery to the bronchus at each lung hilum is described by **RALS**—**R**ight **A**nterior; **L**eft **S**uperior. Carina is posterior to ascending aorta and anteromedial to descending aorta **A**.

Right lung is a more common site for inhaled foreign bodies because right main stem bronchus is wider, more vertical, and shorter than the left. If you aspirate a peanut:

- While upright—enters basal segments of right lower lobe. Preferentially on right, but bilateral basal segments can be involved.
- While supine—enters posterior segment of right upper lobe. Preferentially on right side.



Diaphragm structures



Structures perforating diaphragm:

- At T8: IVC, right phrenic nerve
- At T10: esophagus, vagus (CN 10; 2 trunks)
- At T12: aorta (red), thoracic duct (white), azygos vein (blue) (“At **T-1-2** it’s the **red, white, and blue**”)

Diaphragm is innervated by C3, 4, and 5 (phrenic nerve). Pain from diaphragm irritation (eg, air, blood, or pus in peritoneal cavity) can be referred to shoulder (C5) and trapezius ridge (C3, 4).

Number of letters = T level:

T8: vena cava

T10: “oesophagus”

T12: aortic hiatus

I (IVC) **ate** (8) **ten** (10) **eggs** (esophagus) **at** (aorta) **twelve** (12).

C3, 4, 5 keeps the diaphragm **alive**.

Other bifurcations:

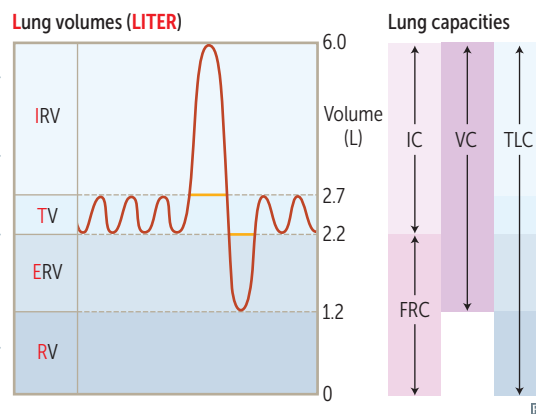
- The common carotid **bifourcates** at C4.
- The trachea **bifourcates** at T4.
- The abdominal aorta **bifourcates** at L4.

► RESPIRATORY—PHYSIOLOGY

Lung volumes

Note: a **capacity** is a sum of ≥ 2 physiologic volumes.

Inspiratory reserve volume	Air that can still be breathed in after normal inspiration
Tidal volume	Air that moves into lung with each quiet inspiration, typically 500 mL
Expiratory reserve volume	Air that can still be breathed out after normal expiration
Residual volume	Air in lung after maximal expiration; RV and any lung capacity that includes RV cannot be measured by spirometry
Inspiratory capacity	IRV + TV Air that can be breathed in after normal exhalation
Functional residual capacity	RV + ERV Volume of gas in lungs after normal expiration
Vital capacity	TV + IRV + ERV Maximum volume of gas that can be expired after a maximal inspiration
Total lung capacity	IRV + TV + ERV + RV Volume of gas present in lungs after a maximal inspiration

**Determination of physiologic dead space**

$$V_D = V_T \times \frac{P_{aCO_2} - P_{E}CO_2}{P_{aCO_2}}$$

V_D = physiologic dead space = anatomic dead space of conducting airways plus alveolar dead space; apex of healthy lung is largest contributor of alveolar dead space. Volume of inspired air that does not take part in gas exchange.

V_T = tidal volume.

P_{aCO_2} = arterial PCO_2 .

$P_{E}CO_2$ = expired air PCO_2 .

P_{aCO_2} , P_{aCO_2} , $P_{E}CO_2$, P_{aCO_2} (refers to order of variables in equation)

Physiologic dead space—approximately equivalent to anatomic dead space in normal lungs. May be greater than anatomic dead space in lung diseases with $\dot{V}/\dot{Q}$ defects.

Pathologic dead space—when part of the respiratory zone becomes unable to perform gas exchange. Ventilated but not perfused.

Ventilation

$$V_A = V_E - V_D$$

Minute ventilation (V_E)	Total volume of gas entering lungs per minute $V_E = V_T \times RR$
Alveolar ventilation (V_A)	Volume of gas per unit time that reaches alveoli $V_A = (V_T - V_D) \times RR$

Normal values:

Respiratory rate (RR) = 12–20 breaths/min

V_T = 500 mL/breath

V_D = 150 mL/breath

Lung and chest wall

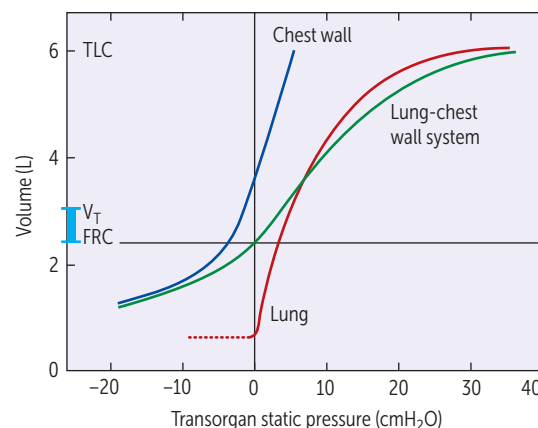
Elastic recoil—tendency for lungs to collapse inward and chest wall to spring outward. At FRC, inward pull of lung is balanced by outward pull of chest wall, and system pressure is atmospheric.

Elastic properties of both chest wall and lungs determine their combined volume.

At FRC, airway and alveolar pressures are 0, and intrapleural pressure is negative (prevents atelectasis). PVR is at minimum.

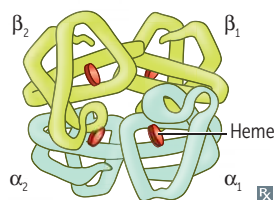
Compliance—change in lung volume for a change in pressure; expressed as $\Delta V/\Delta P$ and is inversely proportional to wall stiffness. High compliance = lung easier to fill (emphysema, normal aging), lower compliance = lung harder to fill (pulmonary fibrosis, pneumonia, NRDS, pulmonary edema). Surfactant increases compliance.

Hysteresis—lung inflation curve follows a different curve than the lung deflation curve due to need to overcome surface tension forces in inflation.



Compliant lungs **comply** (cooperate) and fill easily with air.

Hemoglobin



Hemoglobin (Hb) is composed of 4 polypeptide subunits (2 α and 2 β) and exists in 2 forms:

- Deoxygenated form has low affinity for O₂, thus promoting release/unloading of O₂.
- Oxygenated form has high affinity for O₂ (300 $\times$). Hb exhibits positive cooperativity and negative allosterity.

↑ Cl⁻, H⁺, CO₂, 2,3-BPG, and temperature favor taut form over relaxed form (shifts dissociation curve right → ↑ O₂ unloading).

Fetal Hb (2 α and 2 γ subunits) has a higher affinity for O₂ than adult Hb, driving diffusion of oxygen across the placenta from mother to fetus. ↑ O₂ affinity results from ↓ affinity of HbF for 2,3-BPG.

Hemoglobin acts as buffer for H⁺ ions.

Hemoglobin modifications

Methemoglobin

Oxidized form of Hb (ferric, Fe^{3+}) that does not bind O_2 as readily, but has $\uparrow$ affinity for cyanide.
 Iron in Hb is normally in a reduced state (ferrous, Fe^{2+}).
 Methemoglobinemia may present with cyanosis and chocolate-colored blood.
 Induced methemoglobinemia (using nitrites, followed by thiosulfate) may be used to treat cyanide poisoning.

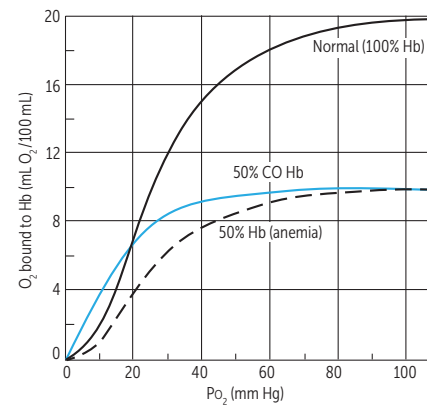
Methemoglobinemia can be treated with **methylene blue** and vitamin C.

Nitrites (eg, from dietary intake or polluted/high altitude water sources) and benzocaine cause poisoning by oxidizing Fe^{2+} to Fe^{3+} .

Fe^{2+} binds O_2 .

Carboxyhemoglobin

Form of Hb bound to CO in place of O_2 .
 Causes $\downarrow$ oxygen-binding capacity with left shift in oxygen-hemoglobin dissociation curve.
 $\downarrow$ O_2 unloading in tissues.
 CO binds competitively to Hb and with 200 $\times$ greater affinity than O_2 .
 CO poisoning can present with headaches, dizziness, and cherry red skin. May be caused by fires, car exhaust, or gas heaters. Treat with 100% O_2 and hyperbaric O_2 .



Oxygen-hemoglobin dissociation curve

Sigmoidal shape due to positive cooperativity (ie, tetrameric Hb molecule can bind 4 O_2 molecules and has higher affinity for each subsequent O_2 molecule bound). Myoglobin is monomeric and thus does not show positive cooperativity; curve lacks sigmoidal appearance.

When curve shifts to the right, $\downarrow$ affinity of Hb for O_2 (facilitates unloading of O_2 to tissue).

An $\uparrow$ in all factors (including H^+) causes a shift of the curve to the right.

A $\downarrow$ in all factors (including H^+) causes a left shift $\rightarrow$ $\downarrow$ O_2 unloading $\rightarrow$ renal hypoxia $\rightarrow$ $\uparrow$ EPO synthesis $\rightarrow$ compensatory erythrocytosis. **Lower = Left**.

Fetal Hb has higher affinity for O_2 than adult Hb (due to low affinity for 2,3-BPG), so its dissociation curve is shifted left.

Right shift—ACE BATs right handed:

Acid

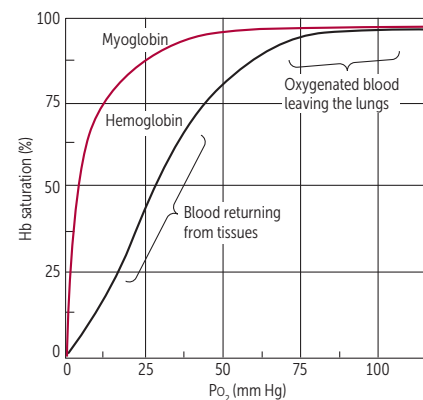
C O_2

Exercise

2,3-BPG

Altitude

Temperature



Oxygen content of blood

$$O_2 \text{ content} = (1.34 \times Hb \times SaO_2) + (0.003 \times PaO_2)$$

Hb = hemoglobin level

SaO₂ = arterial O₂ saturation

PaO₂ = partial pressure of O₂ in arterial blood

Normally 1 g Hb can bind 1.34 mL O₂; normal Hb amount in blood is 15 g/dL.

O₂ binding capacity ≈ 20.1 mL O₂/dL blood.

With ↓ Hb there is ↓ O₂ content of arterial blood, but no change in O₂ saturation and PaO₂.

O₂ delivery to tissues = cardiac output × O₂ content of blood.

	Hb concentration	% O ₂ sat of Hb	Dissolved O ₂ (PaO ₂)	Total O ₂ content
CO poisoning	Normal	↓ (CO competes with O ₂)	Normal	↓
Anemia	↓	Normal	Normal	↓
Polycythemia	↑	Normal	Normal	↑

Pulmonary circulation

Normally a low-resistance, high-compliance system. PO₂ and PCO₂ exert opposite effects on pulmonary and systemic circulation. A ↓ in PaO₂ causes a hypoxic vasoconstriction that shifts blood away from poorly ventilated regions of lung to well-ventilated regions of lung.

Perfusion limited—O₂ (normal health), CO₂, N₂O. Gas equilibrates early along the length of the capillary. Diffusion can be ↑ only if blood flow ↑.

Diffusion limited—O₂ (emphysema, fibrosis), CO. Gas does not equilibrate by the time blood reaches the end of the capillary.

A consequence of pulmonary hypertension is cor pulmonale and subsequent right ventricular failure (jugular venous distention, edema, hepatomegaly).

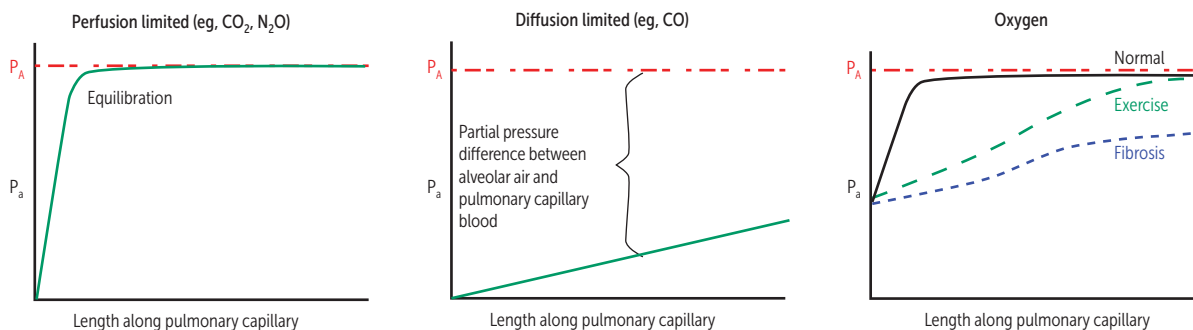
Diffusion: $\dot{V}_{\text{gas}} = A \times D_k \times \frac{P_1 - P_2}{T}$ where

A = area, T = alveolar wall thickness,

D_k = diffusion coefficient of gas, P₁ – P₂ = difference in partial pressures.

- A ↓ in emphysema.
- T ↑ in pulmonary fibrosis.

D_{LCO} is the extent to which CO, a surrogate for O₂, passes from air sacs of lungs into blood.



P_a = partial pressure of gas in pulmonary capillary blood
 P_A = partial pressure of gas in alveolar air

Pulmonary vascular resistance

$$PVR = \frac{P_{\text{pulm artery}} - P_{\text{L atrium}}}{\text{cardiac output}}$$

Remember: $\Delta P = Q \times R$, so $R = \Delta P / Q$

$$R = 8\eta l / \pi r^4$$

R = resistance

$P_{\text{pulm artery}}$ = pressure in pulmonary artery

$P_{\text{L atrium}} \approx$ pulmonary capillary wedge pressure

η = viscosity of blood; l = vessel length;

r = vessel radius

Alveolar gas equation

$$P_{\text{AO}_2} = P_{\text{IO}_2} - \frac{P_{\text{aCO}_2}}{R}$$

$$\approx 150 \text{ mm Hg}^a - \frac{P_{\text{aCO}_2}}{0.8}$$

^aAt sea level breathing room air

P_{AO_2} = alveolar P_{O_2} (mm Hg)

P_{IO_2} = P_{O_2} in inspired air (mm Hg)

P_{aCO_2} = arterial P_{CO_2} (mm Hg)

R = respiratory quotient = CO_2 produced/ O_2 consumed

A-a gradient = $P_{\text{AO}_2} - P_{\text{aO}_2}$. Normal range = 10–15 mm Hg

↑ A-a gradient may occur in hypoxemia; causes include shunting, $\dot{V}/\dot{Q}$ mismatch, fibrosis (impairs diffusion)

Oxygen deprivation

Hypoxia (↓ O_2 delivery to tissue)	Hypoxemia (↓ P_{aO_2})	Ischemia (loss of blood flow)
↓ cardiac output - Hypoxemia - Anemia - CO poisoning	Normal A-a gradient - High altitude - Hypoventilation (eg, opioid use) ↑ A-a gradient $\dot{V}/\dot{Q}$ mismatch - Diffusion limitation (eg, fibrosis) - Right-to-left shunt	Impeded arterial flow ↓ venous drainage

Ventilation/perfusion mismatch

Ideally, ventilation is matched to perfusion (ie, $\dot{V}/\dot{Q} = 1$) for adequate gas exchange.

Lung zones:

- $\dot{V}/\dot{Q}$ at apex of lung = 3 (wasted ventilation)
- $\dot{V}/\dot{Q}$ at base of lung = 0.6 (wasted perfusion)

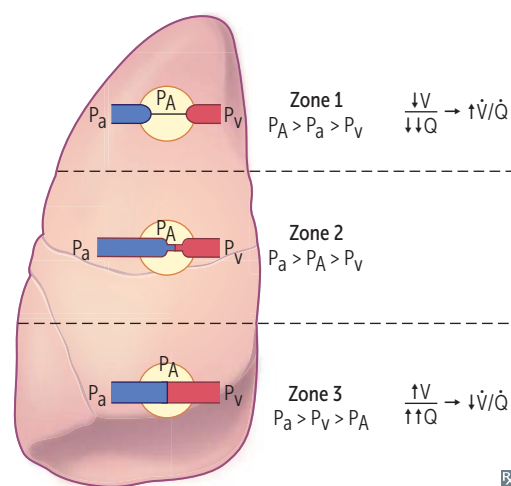
Both ventilation and perfusion are greater at the base of the lung than at the apex of the lung.

With exercise (↑ cardiac output), there is vasodilation of apical capillaries → $\dot{V}/\dot{Q}$ ratio approaches 1.

Certain organisms that thrive in high O_2 (eg, TB) flourish in the apex.

$\dot{V}/\dot{Q} = 0$ = “**o**irway” obstruction (shunt). In shunt, 100% O_2 does not improve P_{aO_2} (eg, foreign body aspiration).

$\dot{V}/\dot{Q} = \infty$ = **blo**od flow obstruction (physiologic dead space). Assuming < 100% dead space, 100% O_2 improves P_{aO_2} (eg, pulmonary embolus).



Carbon dioxide transport

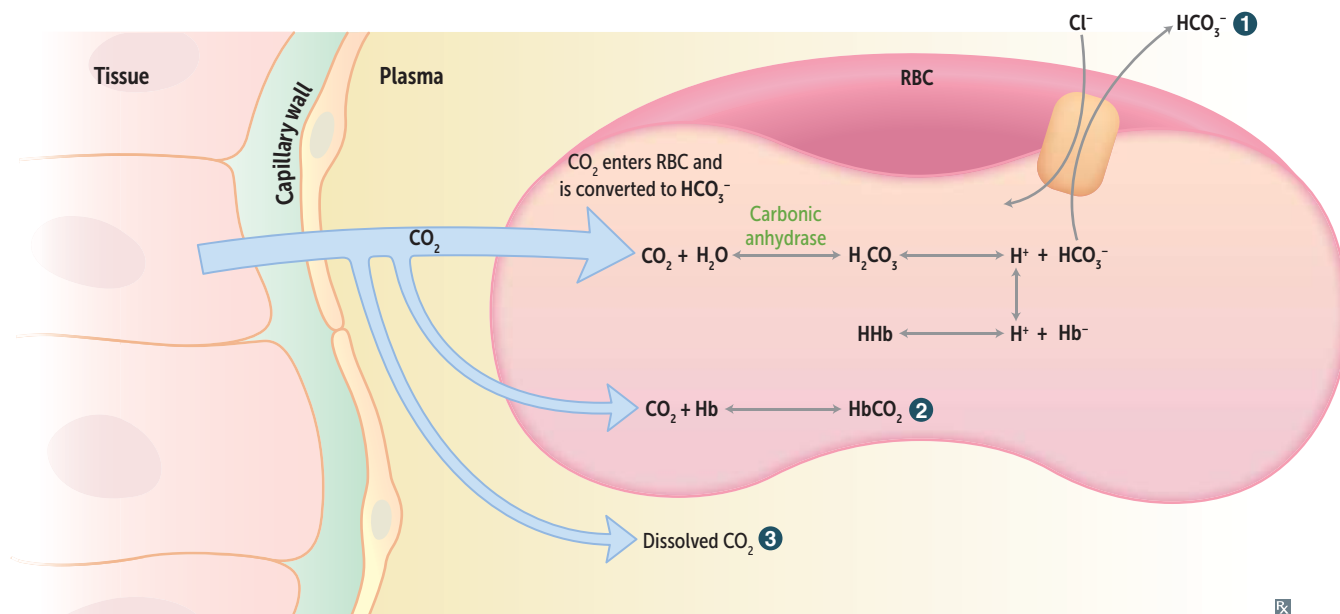
CO₂ is transported from tissues to lungs in 3 forms:

- ❶ HCO₃⁻ (70%).
- ❷ Carbaminohemoglobin or HbCO₂ (21–25%). CO₂ bound to Hb at N-terminus of globin (not heme). CO₂ binding favors taut form (O₂ unloaded).
- ❸ Dissolved CO₂ (5–9%).

In lungs, oxygenation of Hb promotes dissociation of H⁺ from Hb. This shifts equilibrium toward CO₂ formation; therefore, CO₂ is released from RBCs (Haldane effect).

In peripheral tissue, ↑ H⁺ from tissue metabolism shifts curve to right, unloading O₂ (Bohr effect).

Majority of blood CO₂ is carried as HCO₃⁻ in the plasma.

**Response to high altitude**

↓ atmospheric oxygen (PO₂) → ↓ PaO₂ → ↑ ventilation → ↓ PaCO₂ → respiratory alkalosis → altitude sickness.

Chronic ↑ in ventilation.

↑ erythropoietin → ↑ hematocrit and Hb (chronic hypoxia).

↑ 2,3-BPG (binds to Hb so that Hb releases more O₂).

Cellular changes (↑ mitochondria).

↑ renal excretion of HCO₃⁻ to compensate for respiratory alkalosis (can augment with acetazolamide).

Chronic hypoxic pulmonary vasoconstriction results in pulmonary hypertension and RVH.

Response to exercise

↑ CO₂ production.

↑ O₂ consumption.

↑ ventilation rate to meet O₂ demand.

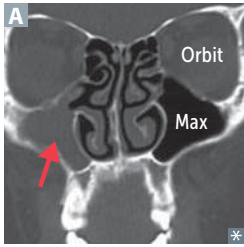
$\dot{V}/\dot{Q}$ ratio from apex to base becomes more uniform.

↑ pulmonary blood flow due to ↑ cardiac output.

↓ pH during strenuous exercise (2° to lactic acidosis).

No change in PaO₂ and PaCO₂, but ↑ in venous CO₂ content and ↓ in venous O₂ content.

► RESPIRATORY—PATHOLOGY

Rhinosinusitis

Obstruction of sinus drainage into nasal cavity → inflammation and pain over affected area (typically maxillary sinuses, filled with fluid on the right in **A**, which drain into the middle meatus, in adults).

Most common acute cause is viral URI; may cause superimposed bacterial infection, most commonly *S pneumoniae*, *H influenzae*, *M catarrhalis*.

Epistaxis

Nose bleed. Most commonly occurs in anterior segment of nostril (Kiesselbach plexus). Life-threatening hemorrhages occur in posterior segment (sphenopalatine artery, a branch of maxillary artery). Common causes include foreign body, trauma, allergic rhinitis, and nasal angiofibromas.

Head and neck cancer

Mostly squamous cell carcinoma. Risk factors include tobacco, alcohol, HPV-16 (oropharyngeal), EBV (nasopharyngeal). Field cancerization: carcinogen damages wide mucosal area → multiple tumors that develop independently after exposure.

Deep venous thrombosis

Blood clot within a deep vein → swelling, redness **A**, warmth, pain. Predisposed by Virchow triad (**SHE**):

- **S**tasis (eg, post-op, long drive/flight)
- **H**ypercoagulability (eg, defect in coagulation cascade proteins, such as factor V Leiden)
- **E**ndothelial damage (exposed collagen triggers clotting cascade)

D-dimer lab test used clinically to rule out DVT (high sensitivity, low specificity).

Most pulmonary emboli arise from proximal deep veins of lower extremity.

Use unfractionated heparin or low-molecular-weight heparins (eg, enoxaparin) for prophylaxis and acute management.

Use oral anticoagulants (eg, warfarin, rivaroxaban) for treatment (long-term prevention).

Imaging test of choice is compression ultrasound with Doppler.

Pulmonary emboli

$\dot{V}/\dot{Q}$ mismatch, hypoxemia, respiratory alkalosis. Sudden-onset dyspnea, pleuritic chest pain, tachypnea, tachycardia. Large emboli or saddle embolus **A** may cause sudden death.

Lines of Zahn are interdigitating areas of pink (platelets, fibrin) and red (RBCs) found only in thrombi formed before death; help distinguish pre- and postmortem thrombi **B**.

Types: **F**at, **A**ir, **T**hrombus, **B**acteria, **A**mniotic fluid, **T**umor.

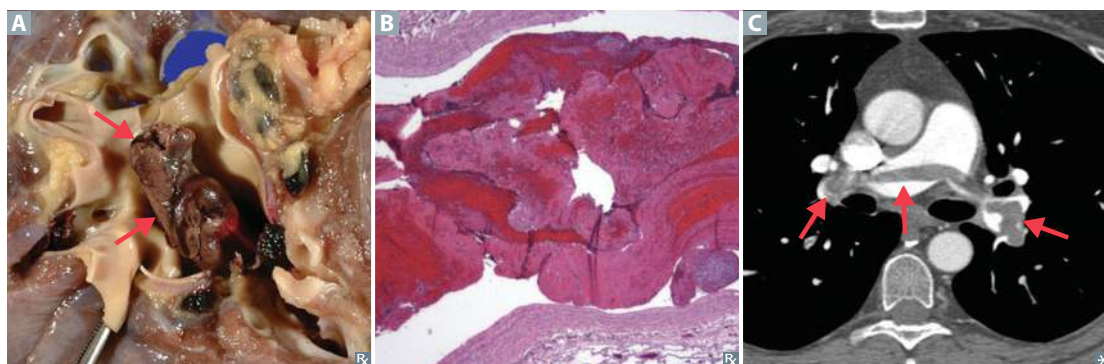
Fat emboli—associated with long bone fractures and liposuction; classic triad of hypoxemia, neurologic abnormalities, petechial rash.

Amniotic fluid emboli—can lead to DIC, especially postpartum.

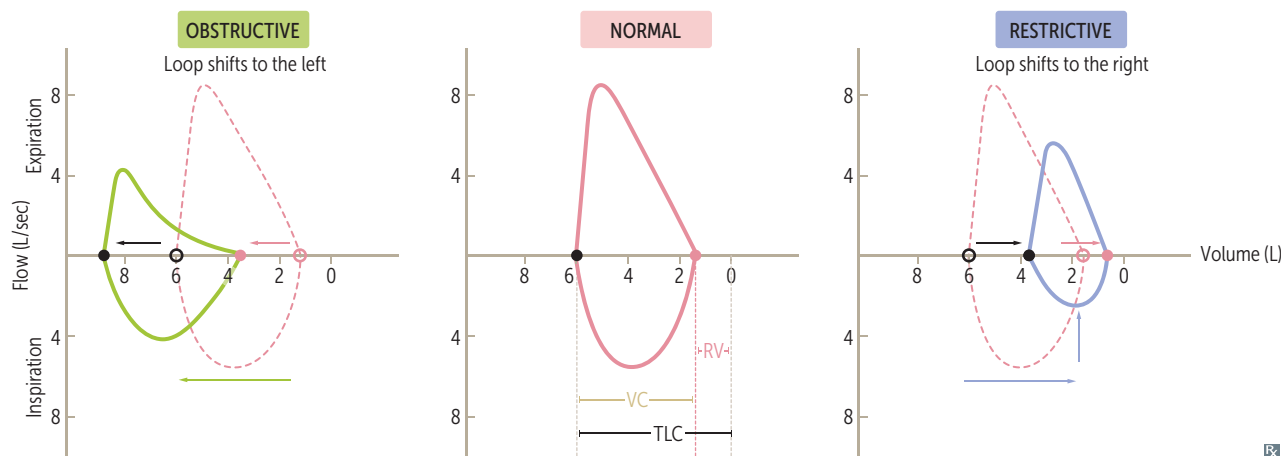
Air emboli—nitrogen bubbles precipitate in ascending divers (caisson disease, decompression sickness); treat with hyperbaric O_2 ; or, can be iatrogenic 2° to invasive procedures (eg, central line placement).

CT pulmonary angiography is imaging test of choice for PE (look for filling defects) **C**.

An embolus moves like a **FAT BAT**.

**Flow volume loops**

Obstructive lung volumes > normal ($\uparrow$ TLC, $\uparrow$ FRC, $\uparrow$ RV); restrictive lung volumes < normal. In obstructive, FEV_1 is more dramatically reduced compared with FVC $\rightarrow$ decreased FEV_1/FVC ratio. In restrictive, FVC is more reduced or close to same compared with $FEV_1 \rightarrow$ increased or normal FEV_1/FVC ratio.



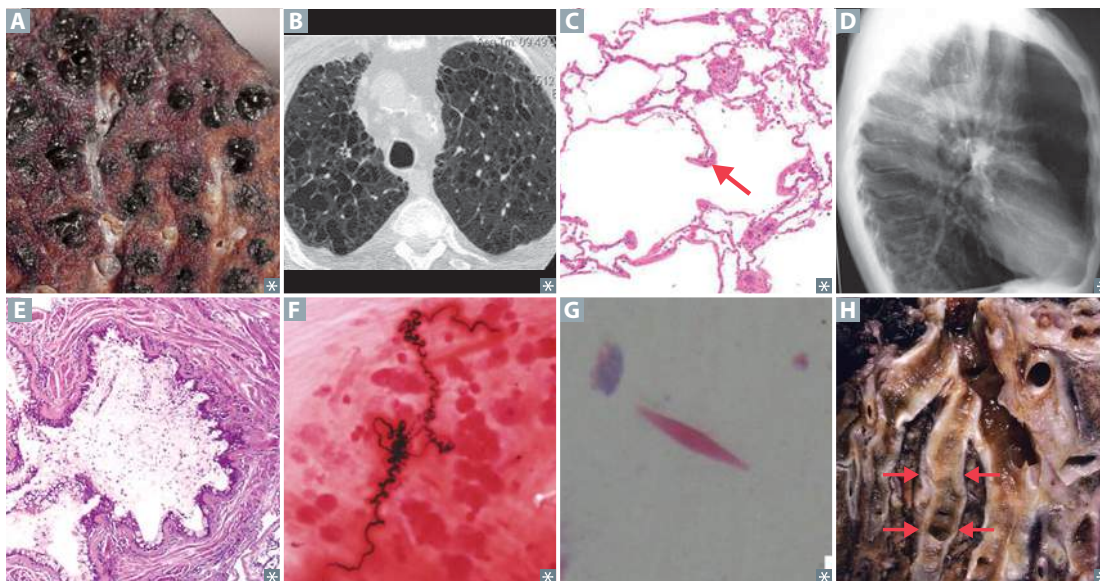
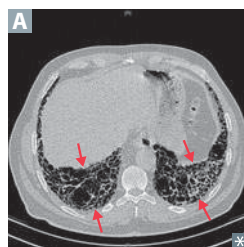
Obstructive lung diseases

Obstruction of air flow → air trapping in lungs. Airways close prematurely at high lung volumes → ↑ RV and ↑ FRC, ↑ TLC. PFTs: ↓↓ FEV₁, ↓ FVC → ↓ FEV₁/FVC ratio (hallmark), V/Q mismatch. Chronic, hypoxic pulmonary vasoconstriction can lead to cor pulmonale. Chronic obstructive pulmonary disease (COPD) includes chronic bronchitis and emphysema.

TYPE	PRESENTATION	PATHOLOGY	OTHER
Chronic bronchitis ("blue bloater")	Findings: wheezing, crackles, cyanosis (hypoxemia due to shunting), dyspnea, CO ₂ retention, 2° polycythemia.	Hypertrophy and hyperplasia of mucus-secreting glands in bronchi → Reid index (thickness of mucosal gland layer to thickness of wall between epithelium and cartilage) > 50%.	Diagnostic criteria: productive cough for > 3 months in a year for > 2 consecutive years.
Emphysema ("pink puffer")	Centriacinar—associated with smoking A B . Frequently in upper lobes (smoke rises up). Panacinar—associated with α ₁ -antitrypsin deficiency. Frequently in lower lobes.	Enlargement of air spaces ↓ recoil, ↑ compliance, ↓ D _{LCO} from destruction of alveolar walls (arrow in C). ↑ elastase activity → ↑ loss of elastic fibers → ↑ lung compliance.	CXR: ↑ AP diameter, flattened diaphragm, ↑ lung field lucency. Barrel-shaped chest D . Exhalation through pursed lips to increase airway pressure and prevent airway collapse.
Asthma	Findings: cough, wheezing, tachypnea, dyspnea, hypoxemia, ↓ inspiratory/expiratory ratio, pulsus paradoxus, mucus plugging E . Triggers: viral URIs, allergens, stress. Diagnosis supported by spirometry and methacholine challenge.	Bronchial hyperresponsiveness → reversible bronchoconstriction. Smooth muscle hypertrophy and hyperplasia, Curschmann spirals F (shed epithelium forms whorled mucous plugs), and Charcot-Leyden crystals G (eosinophilic, hexagonal, double-pointed, needle-like crystals formed from breakdown of eosinophils in sputum).	Aspirin-induced asthma: COX inhibition → leukotriene overproduction → airway constriction. Associated with nasal polyps.

Obstructive lung diseases (continued)

TYPE	PRESENTATION	PATHOLOGY	OTHER
Bronchiectasis	Findings: purulent sputum, recurrent infections, hemoptysis, digital clubbing.	Chronic necrotizing infection of bronchi → permanently dilated airways.	Associated with bronchial obstruction, poor ciliary motility (eg, smoking, Kartagener syndrome), cystic fibrosis H , allergic bronchopulmonary aspergillosis.

**Restrictive lung diseases**

Restricted lung expansion causes ↓ lung volumes (↓ FVC and TLC). PFTs: FEV_1/FVC ratio $\geq 80\%$. Patient presents with short, shallow breaths.

Types:

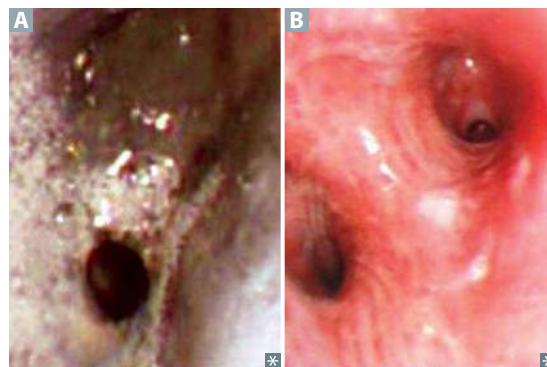
- Poor breathing mechanics (extrapulmonary, peripheral hypoventilation, normal A-a gradient):
 - Poor muscular effort—polio, myasthenia gravis, Guillain-Barré syndrome
 - Poor structural apparatus—scoliosis, morbid obesity
- Interstitial lung diseases (pulmonary ↓ diffusing capacity, ↑ A-a gradient):
 - Pneumoconioses (eg, coal workers' pneumoconiosis, silicosis, asbestosis)
 - Sarcoidosis: bilateral hilar lymphadenopathy, noncaseating granuloma; ↑ ACE and Ca^{2+}
 - Idiopathic pulmonary fibrosis **A** (repeated cycles of lung injury and wound healing with ↑ collagen deposition, “honeycomb” lung appearance and digital clubbing)
 - Granulomatosis with polyangiitis (Wegener)
 - Pulmonary Langerhans cell histiocytosis (eosinophilic granuloma)
 - Hypersensitivity pneumonitis
 - Drug toxicity (bleomycin, busulfan, amiodarone, methotrexate)

Hypersensitivity pneumonitis—mixed type III/IV hypersensitivity reaction to environmental antigen. Causes dyspnea, cough, chest tightness, headache. Often seen in farmers and those exposed to birds. Reversible in early stages if stimulus is avoided.

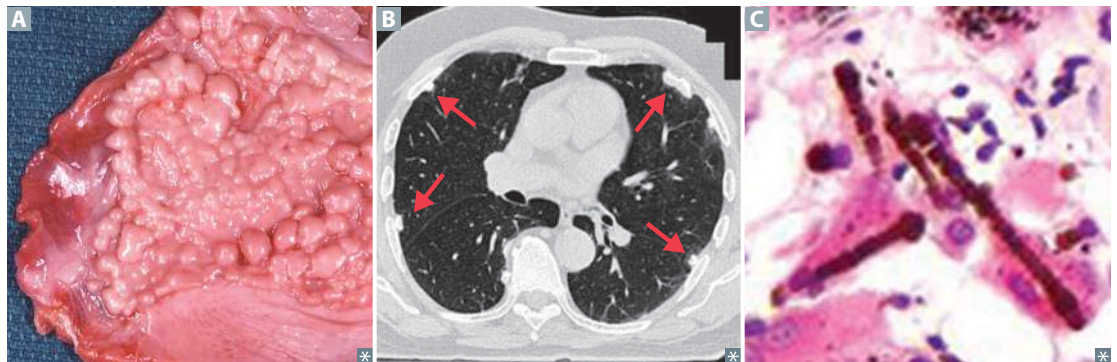
Inhalation injury and sequelae

Pulmonary complication associated with smoke and fire. Caused by heat, particulates ($< 1 \mu\text{m}$ diameter), or irritants (eg, NH_3) $\rightarrow$ chemical tracheobronchitis, edema, pneumonia, ARDS. Many patients present 2° to burns, CO inhalation, cyanide poisoning, or arsenic poisoning.

Bronchoscopy shows severe edema, congestion of bronchus, and soot deposition (**A**, 18 hours after inhalation injury; **B**, resolution at 11 days after injury).

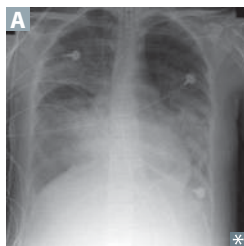


Pneumoconioses	Coal workers' pneumoconiosis, silicosis, and asbestosis → ↑ risk of cor pulmonale, cancer, and Caplan syndrome (rheumatoid arthritis and pneumoconioses with intrapulmonary nodules).	Asbestos is from the roof (was common in insulation), but affects the base (lower lobes). Silica and coal are from the base (earth), but affect the roof (upper lobes).
Asbestosis	Associated with shipbuilding, roofing, plumbing. "Ivory white," calcified, supradiaphragmatic A and pleural B plaques are pathognomonic of asbestosis. Risk of bronchogenic carcinoma > risk of mesothelioma.	Affects lower lobes. Asbestos (ferruginous) bodies are golden-brown fusiform rods resembling dumbbells C , found in alveolar sputum sample, visualized using Prussian blue stain, often obtained by bronchoalveolar lavage. ↑ risk of pleural effusions.
Berylliosis	Associated with exposure to beryllium in aerospace and manufacturing industries. Granulomatous (noncaseating) on histology and therefore occasionally responsive to steroids.	Affects upper lobes.
Coal workers' pneumoconiosis	Prolonged coal dust exposure → macrophages laden with carbon → inflammation and fibrosis. Also known as black lung disease.	Affects upper lobes. Anthracosis —asymptomatic condition found in many urban dwellers exposed to sooty air.
Silicosis	Associated with foundries, sandblasting, mines. Macrophages respond to silica and release fibrogenic factors, leading to fibrosis. It is thought that silica may disrupt phagolysosomes and impair macrophages, increasing susceptibility to TB.	Affects upper lobes. "Eggshell" calcification of hilar lymph nodes on CXR.

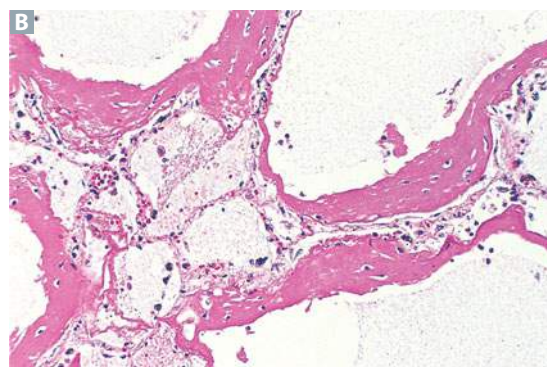


Mesothelioma	Malignancy of the pleura associated with asbestosis. May result in hemorrhagic pleural effusion (exudative), pleural thickening.	Psammoma bodies seen on histology. Cytokeratin and calretinin ⊕ in almost all mesotheliomas, ⊖ in most carcinomas. Smoking not a risk factor.
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Acute respiratory distress syndrome



Diagnosis of exclusion characterized by respiratory failure within 1 week of alveolar insult, bilateral lung opacities, $\downarrow$ $\text{PaO}_2/\text{FiO}_2 < 300$ (hypoxemia due to $\uparrow$ intrapulmonary shunting and diffusion abnormalities), no evidence of HF/fluid overload. Many causes and associations, including sepsis, pancreatitis, pneumonia, aspiration, trauma, shock. Endothelial damage $\rightarrow$ $\uparrow$ alveolar capillary permeability $\rightarrow$ protein-rich leakage into alveoli $\rightarrow$ diffuse alveolar damage and noncardiogenic pulmonary edema (normal PCWP) **A**. Results in formation of intra-alveolar hyaline membranes **B**. Initial damage due to release of neutrophilic substances toxic to alveolar wall and pulmonary capillary endothelial cells, activation of coagulation cascade, and oxygen-derived free radicals. Management: mechanical ventilation with low tidal volumes, address underlying cause.



Sleep apnea

Repeated cessation of breathing > 10 seconds during sleep $\rightarrow$ disrupted sleep $\rightarrow$ daytime somnolence. Diagnosis confirmed by sleep study. Normal PaO_2 during the day. Nocturnal hypoxia $\rightarrow$ systemic/pulmonary hypertension, arrhythmias (atrial fibrillation/flutter), sudden death. Hypoxia $\rightarrow$ $\uparrow$ EPO release $\rightarrow$ $\uparrow$ erythropoiesis.

Obstructive sleep apnea

Respiratory effort against airway obstruction. Associated with obesity, loud snoring. Caused by excess parapharyngeal tissue in adults, adenotonsillar hypertrophy in children. Treatment: weight loss, CPAP, surgery.

Central sleep apnea

No respiratory effort due to **CNS** injury/toxicity, HF, opioids. May be associated with Cheyne-Stokes respiration. Treat with positive airway pressure.

Obesity hypoventilation syndrome

Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) $\rightarrow$ hypoventilation $\uparrow$ PaCO_2 during waking hours (retention); $\downarrow$ PaO_2 and $\uparrow$ PaCO_2 during sleep. Also known as Pickwickian syndrome.

Pulmonary hypertension

Normal mean pulmonary artery pressure = 10–14 mm Hg; pulmonary hypertension $\geq$ 25 mm Hg at rest. Results in arteriosclerosis, medial hypertrophy, intimal fibrosis of pulmonary arteries, plexiform lesions. Course: severe respiratory distress $\rightarrow$ cyanosis and RVH $\rightarrow$ death from decompensated cor pulmonale.

ETIOLOGIES

Pulmonary arterial hypertension	Idiopathic PAH. Heritable PAH—often due to an inactivating mutation in <i>BMPR2</i> gene (normally inhibits vascular smooth muscle proliferation); poor prognosis. Other causes include drugs (eg, amphetamines, cocaine), connective tissue disease, HIV infection, portal hypertension, congenital heart disease, schistosomiasis.
Left heart disease	Causes include systolic/diastolic dysfunction and valvular disease (eg, mitral lung).
Lung diseases or hypoxia	Destruction of lung parenchyma (eg, COPD), lung inflammation/fibrosis (eg, interstitial lung diseases), hypoxemic vasoconstriction (eg, obstructive sleep apnea, living in high altitude).
Chronic thromboembolic	Recurrent microthrombi $\rightarrow$ $\downarrow$ cross-sectional area of pulmonary vascular bed.
Multifactorial	Causes include hematologic, systemic, and metabolic disorders.

Lung—physical findings

ABNORMALITY	BREATH SOUNDS	PERCUSSION	FREMITUS	TRACHEAL DEVIATION
Pleural effusion	$\downarrow$	Dull	$\downarrow$	— or away from side of lesion (if large)
Atelectasis (bronchial obstruction)	$\downarrow$	Dull	$\downarrow$	Toward side of lesion
Simple pneumothorax	$\downarrow$	Hyperresonant	$\downarrow$	—
Tension pneumothorax	$\downarrow$	Hyperresonant	$\downarrow$	Away from side of lesion
Consolidation (lobar pneumonia, pulmonary edema)	Bronchial breath sounds; late inspiratory crackles, egophony, bronchophony, whispered pectoriloquy	Dull	$\uparrow$	—

Pleural effusions

Excess accumulation of fluid **A** between pleural layers → restricted lung expansion during inspiration. Can be treated with thoracentesis to remove/reduce fluid **B**.

Transudate

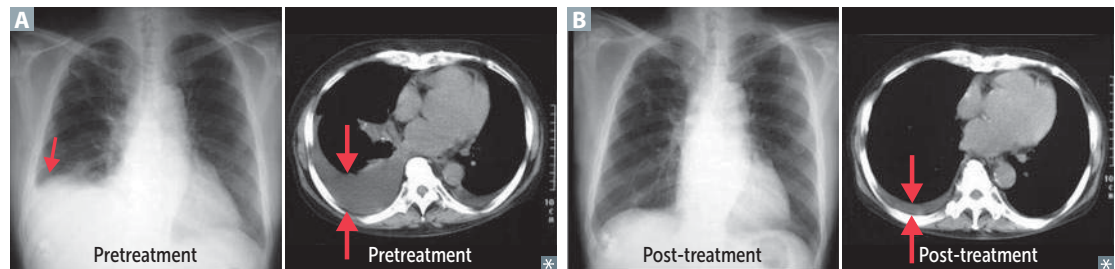
↓ protein content. Due to ↑ hydrostatic pressure (eg, HF) or ↓ oncotic pressure (eg, nephrotic syndrome, cirrhosis).

Exudate

↑ protein content, cloudy. Due to malignancy, pneumonia, collagen vascular disease, trauma (occurs in states of ↑ vascular permeability). Must be drained due to risk of infection.

Lymphatic

Also known as chylothorax. Due to thoracic duct injury from trauma or malignancy. Milky-appearing fluid; ↑ triglycerides.

**Pneumothorax**

Accumulation of air in pleural space **A**. Dyspnea, uneven chest expansion. Chest pain, ↓ tactile fremitus, hyperresonance, and diminished breath sounds, all on the affected side.

Primary spontaneous pneumothorax

Due to rupture of apical subpleural bleb or cysts. Occurs most frequently in tall, thin, young males.

Secondary spontaneous pneumothorax

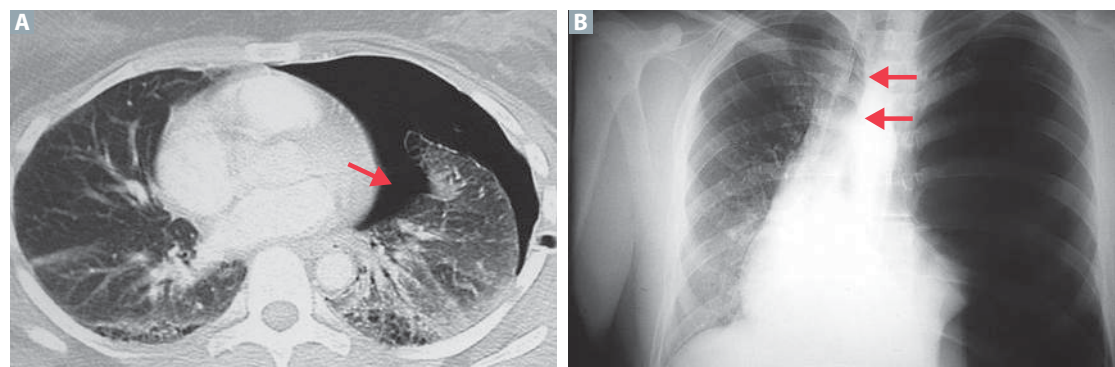
Due to diseased lung (eg, bullae in emphysema, infections), mechanical ventilation with use of high pressures → barotrauma.

Traumatic pneumothorax

Caused by blunt (eg, rib fracture) or penetrating (eg, gunshot) trauma.

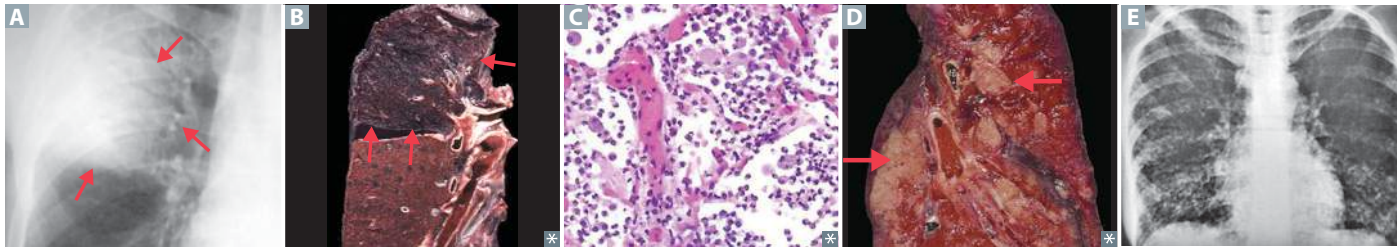
Tension pneumothorax

Can be any of the above. Air enters pleural space but cannot exit. Increasing trapped air → tension pneumothorax. Trachea deviates away from affected lung **B**. Needs immediate needle decompression and chest tube placement.

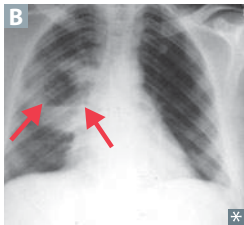


Pneumonia

TYPE	TYPICAL ORGANISMS	CHARACTERISTICS
Lobar	<i>S pneumoniae</i> most frequently, also <i>Legionella</i> , <i>Klebsiella</i>	Intra-alveolar exudate → consolidation A ; may involve entire lobe B or lung.
Bronchopneumonia	<i>S pneumoniae</i> , <i>S aureus</i> , <i>H influenzae</i> , <i>Klebsiella</i>	Acute inflammatory infiltrates C from bronchioles into adjacent alveoli; patchy distribution involving ≥ 1 lobe D .
Interstitial (atypical) pneumonia	<i>Mycoplasma</i> , <i>Chlamydophila pneumoniae</i> , <i>Chlamydia psittaci</i> , <i>Legionella</i> , viruses (RSV, CMV, influenza, adenovirus)	Diffuse patchy inflammation localized to interstitial areas at alveolar walls; diffuse distribution involving ≥ 1 lobe E . Generally follows a more indolent course (“walking” pneumonia).
Cryptogenic organizing pneumonia		Formerly known as bronchiolitis obliterans organizing pneumonia (BOOP). Noninfectious pneumonia characterized by inflammation of bronchioles and surrounding structure. Etiology unknown. Secondary organizing pneumonia caused by chronic inflammatory diseases (eg, rheumatoid arthritis) or medication side effects (eg, amiodarone). ⊖ sputum and blood cultures, no response to antibiotics.

**Natural history of lobar pneumonia**

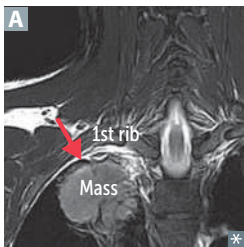
	Congestion	Red hepatization	Gray hepatization	Resolution
DAYS	1–2	3–4	5–7	8+
FINDINGS	Red-purple, partial consolidation of parenchyma Exudate with mostly bacteria	Red-brown, consolidated Exudate with fibrin, bacteria, RBCs, and WBCs	Uniformly gray Exudate full of WBCs and fibrin	Enzymes digest components of exudate

Lung abscess

Localized collection of pus within parenchyma **A**. Caused by aspiration of oropharyngeal contents (especially in patients predisposed to loss of consciousness [eg, alcoholics, epileptics]) or bronchial obstruction (eg, cancer).

Treatment: clindamycin.

Air-fluid levels **B** often seen on CXR. Fluid levels common in cavities; presence suggests cavitation. Due to anaerobes (eg, *Bacteroides*, *Fusobacterium*, *Peptostreptococcus*) or *S aureus*. Lung abscess 2° to aspiration is most often found in right lung. Location depends on patient's position during aspiration.

Pancoast tumor (superior sulcus tumor)

Carcinoma that occurs in the apex of lung **A** may cause Pancoast syndrome by invading cervical sympathetic chain.

Compression of locoregional structures may cause array of findings:

- Recurrent laryngeal nerve → hoarseness
- Stellate ganglion → Horner syndrome (ipsilateral ptosis, miosis, anhidrosis)
- Superior vena cava → SVC syndrome
- Brachiocephalic vein → brachiocephalic syndrome (unilateral symptoms)
- Brachial plexus → sensorimotor deficits

Superior vena cava syndrome

An obstruction of the SVC that impairs blood drainage from the head ("facial plethora"; note blanching after fingertip pressure in **A**), neck (jugular venous distention), and upper extremities (edema). Commonly caused by malignancy (eg, mediastinal mass, Pancoast tumor) and thrombosis from indwelling catheters **B**. Medical emergency. Can raise intracranial pressure (if obstruction is severe) → headaches, dizziness, ↑ risk of aneurysm/rupture of intracranial arteries.



Lung cancer

Leading cause of cancer death.

Presentation: cough, hemoptysis, bronchial obstruction, wheezing, pneumonic “coin” lesion on CXR or noncalcified nodule on CT.

Sites of metastases from lung cancer: adrenals, brain, bone (pathologic fracture), liver (jaundice, hepatomegaly).

In the lung, metastases (usually multiple lesions) are more common than 1° neoplasms. Most often from breast, colon, prostate, and bladder cancer.

SPHERE of complications:

Superior vena cava syndrome

Pancoast tumor

Horner syndrome

Endocrine (paraneoplastic)

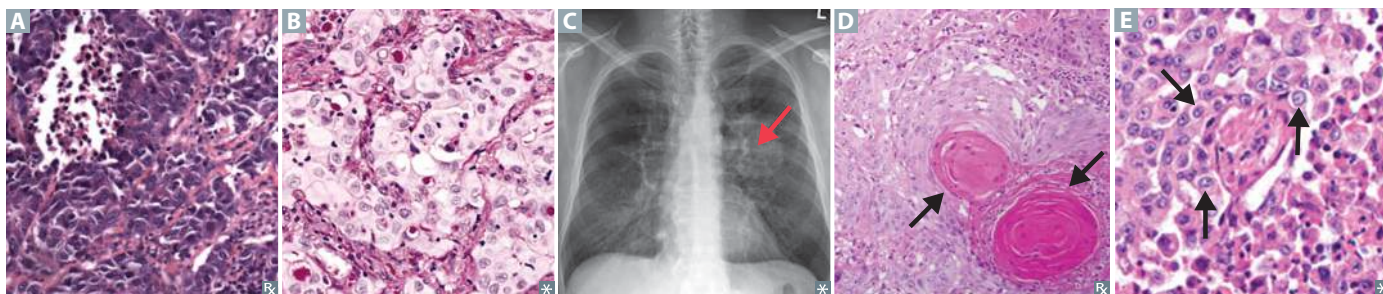
Recurrent laryngeal nerve compression (hoarseness)

Effusions (pleural or pericardial)

Risk factors include smoking, secondhand smoke, radon, asbestos, family history.

Squamous and **S**mall cell carcinomas are **S**entral (central) and often caused by **S**moking.

TYPE	LOCATION	CHARACTERISTICS	HISTOLOGY
Small cell			
Small cell (oat cell) carcinoma	Central	Undifferentiated → very aggressive. May produce ACTH (Cushing syndrome), SIADH , or Antibodies against presynaptic Ca^{2+} channels (Lambert-Eaton myasthenic syndrome) or neurons (paraneoplastic myelitis, encephalitis, subacute cerebellar degeneration). Amplification of <i>myc</i> oncogenes common. Managed with chemotherapy +/- radiation.	Neoplasm of neuroendocrine Kulchitsky cells → small dark blue cells A . Chromogranin A ⊕, neuron-specific enolase ⊕.
Non-small cell			
Adenocarcinoma	Peripheral	Most common lung cancer in nonsmokers and overall (except for metastases). Activating mutations include <i>KRAS</i> , <i>EGFR</i> , and <i>ALK</i> . Associated with hypertrophic osteoarthropathy (clubbing). Bronchioloalveolar subtype (adenocarcinoma in situ): CXR often shows hazy infiltrates similar to pneumonia; better prognosis. Bronchial carcinoid and bronchioloalveolar cell carcinoma have lesser association with smoking.	Glandular pattern on histology, often stains mucin ⊕ B . Bronchioloalveolar subtype: grows along alveolar septa → apparent “thickening” of alveolar walls. Tall, columnar cells containing mucus.
Squamous cell carcinoma	C entral	Hilar mass C arising from bronchus; C avitation; C igarettes; hyper C alcemia (produces PTHrP).	Keratin pearls D and intercellular bridges.
Large cell carcinoma	Peripheral	Highly anaplastic undifferentiated tumor; poor prognosis. Less responsive to chemotherapy; removed surgically. Strong association with smoking.	Pleomorphic giant cells E .
Bronchial carcinoid tumor	—	Excellent prognosis; metastasis rare. Symptoms due to mass effect or carcinoid syndrome (flushing, diarrhea, wheezing).	Nests of neuroendocrine cells; chromogranin A ⊕.



► RESPIRATORY—PHARMACOLOGY

Antihistamines	Reversible inhibitors of H ₁ histamine receptors.	
First generation	Diphenhydramine, dimenhydrinate, chlorpheniramine.	Names contain “-en/-ine” or “-en/-ate.”
CLINICAL USES	Allergy, motion sickness, sleep aid.	
ADVERSE EFFECTS	Sedation, antimuscarinic, anti- α -adrenergic.	
Second generation	Loratadine, fexofenadine, desloratadine, cetirizine.	Names usually end in “-adine.”
CLINICAL USES	Allergy.	
ADVERSE EFFECTS	Far less sedating than 1st generation because of ↓ entry into CNS.	

Guaifenesin Expectorant—thins respiratory secretions; does not suppress cough reflex.

N-acetylcysteine Mucolytic—liquifies mucus in chronic bronchopulmonary diseases (eg, COPD, CF) by disrupting disulfide bonds. Also used as an antidote for acetaminophen overdose.

Dextromethorphan Antitussive (antagonizes NMDA glutamate receptors). Synthetic codeine analog. Has mild opioid effect when used in excess. Naloxone can be given for overdose. Mild abuse potential. May cause serotonin syndrome if combined with other serotonergic agents.

Pseudoephedrine, phenylephrine

MECHANISM	α -adrenergic agonists, used as nasal decongestants.	
CLINICAL USE	Reduce hyperemia, edema, nasal congestion; open obstructed eustachian tubes. Pseudoephedrine also illicitly used to make methamphetamine.	
ADVERSE EFFECTS	Hypertension. Rebound congestion if used more than 4–6 days. Can also cause CNS stimulation/anxiety (pseudoephedrine).	

Pulmonary hypertension drugs

DRUG	MECHANISM	CLINICAL NOTES
BosENtan	Competitively antagonizes EN dothelin-1 receptors → ↓ pulmonary vascular resistance.	Hepatotoxic (monitor LFTs).
Sildenafil	Inhibits PDE-5 → ↑ cGMP → prolonged vasodilatory effect of nitric oxide.	Also used to treat erectile dysfunction.
Epoprostenol, iloprost	PGI ₂ (prostacyclin) with direct vasodilatory effects on pulmonary and systemic arterial vascular beds. Inhibits platelet aggregation.	Side effects: flushing, jaw pain.

Asthma drugs

Bronchoconstriction is mediated by (1) inflammatory processes and (2) parasympathetic tone; therapy is directed at these 2 pathways.

 β_2 -agonists

Albuterol—relaxes bronchial smooth muscle (short acting β_2 -agonist). Used during acute exacerbation.

Salmeterol, formoterol—long-acting agents for prophylaxis. Adverse effects are tremor and arrhythmia.

Inhaled corticosteroids

Fluticasone, budesonide—inhibit the synthesis of virtually all cytokines. Inactivate NF- κ B, the transcription factor that induces production of TNF- α and other inflammatory agents. 1st-line therapy for chronic asthma. May cause oral thrush.

Muscarinic antagonists

Tiotropium, ipratropium—competitively block muscarinic receptors, preventing bronchoconstriction. Also used for COPD. Tiotropium is long acting.

Antileukotrienes

Montelukast, zafirlukast—block leukotriene receptors (CysLT1). Especially good for aspirin-induced and exercise-induced asthma.

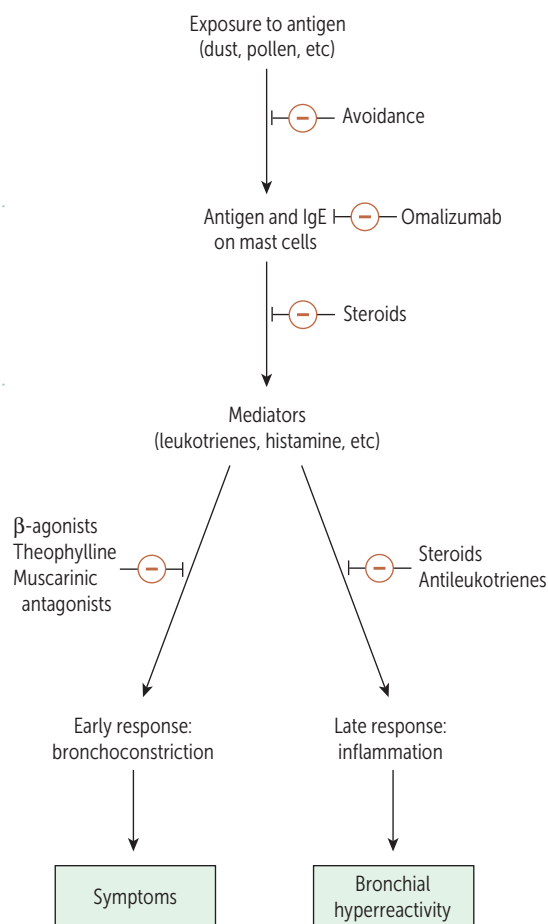
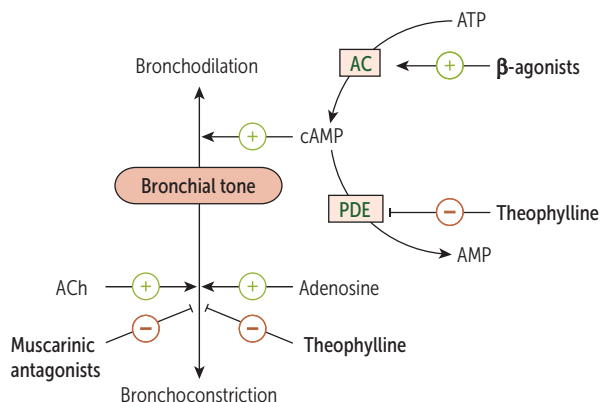
Zileuton—5-lipoxygenase pathway inhibitor. Blocks conversion of arachidonic acid to leukotrienes. Hepatotoxic.

Anti-IgE monoclonal therapy

Omalizumab—binds mostly unbound serum IgE and blocks binding to Fc ϵ RI. Used in allergic asthma with $\uparrow$ IgE levels resistant to inhaled steroids and long-acting β_2 -agonists.

Methylxanthines

Theophylline—likely causes bronchodilation by inhibiting phosphodiesterase $\rightarrow$ $\uparrow$ cAMP levels due to $\downarrow$ cAMP hydrolysis. Usage is limited because of narrow therapeutic index (cardiotoxicity, neurotoxicity); metabolized by cytochrome P-450. Blocks actions of adenosine.

**Methacholine**

Nonselective muscarinic receptor (M_3) agonist. Used in bronchial challenge test to help diagnose asthma.

▶ NOTES